

27 Supersymmetric Gauge Theories

The successful theories of strong, weak, and electromagnetic interactions described in the first two volumes are all gauge theories. In order to see how simple supersymmetry may make contact with reality, we must therefore consider how to construct actions that satisfy both supersymmetry and gauge invariance [1].

27.1 Gauge-Invariant Actions for Chiral Superfields

Consider a set of Abelian or non-Abelian gauge transformations that leave the supersymmetry generator Q invariant. (For simple supersymmetry there is just one Majorana spinor supersymmetry generator, which can only furnish a trivial representation of any semi-simple gauge group.) Each component field in a supermultiplet must transform in the same way under such gauge transformations. In particular, for a left-chiral superfield we have

$$\begin{aligned}\phi_n(x) &\longrightarrow \sum_m \left[\exp \left(i \sum_A t_A \Lambda^A(x) \right) \right]_{nm} \phi_m(x), \\ \psi_{nL}(x) &\longrightarrow \sum_m \left[\exp \left(i \sum_A t_A \Lambda^A(x) \right) \right]_{nm} \psi_{mL}(x), \\ \mathcal{F}_n(x) &\longrightarrow \sum_m \left[\exp \left(i \sum_A t_A \Lambda^A(x) \right) \right]_{nm} \mathcal{F}_m(x),\end{aligned}\tag{27.1.1}$$

where t_A are Hermitian matrices representing the generators of the gauge algebra, and $\Lambda^A(x)$ are real functions of x^μ that parameterize a finite gauge transformation. (We are using the same notation for gauge transformations as in Section 15.1, except that in order to avoid confusion with Dirac indices we label the gauge generators and gauge transformation parameters with the letters A, B , etc. instead of α, β , etc.)

The left-chiral superfield (26.3.11) involves derivatives of some of the component fields, so its transformation is more complicated than Eq. (27.1.1). However, Eq. (26.3.21) shows that the superfield does not involve derivatives if expressed in terms of θ_L and the variable x_+ defined by Eq. (26.3.23). It therefore has the transformation property

$$\Phi_n(x, \theta) \longrightarrow \sum_m \left[\exp \left(i \sum_A t_A \Lambda^A(x_+) \right) \right]_{nm} \Phi_m(x, \theta).\tag{27.1.2}$$

If a term in the action depends only on left-chiral superfields, and not their derivatives or complex conjugates, like the term $\int d^4x [f(\Phi)]_{\mathcal{F}}$ in Eq. (26.3.30), then it (and its complex conjugate) will be invariant under the local transformation (27.1.2) if it is invariant under global transformations with $\Lambda^A(x)$ independent of x^μ . The need for introducing gauge fields in renormalizable theories of chiral superfields arises only in the D -terms, which involve both Φ_n and Φ_n^* . Because the matrices t_A are Hermitian, the Hermitian adjoint of Eq. (27.1.2) is

$$\Phi_n^\dagger(x, \theta) \longrightarrow \sum_m \Phi_m^\dagger(x, \theta) \left[\exp \left(-i \sum_A t_A \Lambda^A(x_+)^* \right) \right]_{mn}.\tag{27.1.3}$$

If it were not for the difference between $\Lambda^A(x_+)^* = \Lambda^A(x_-)$ and $\Lambda^A(x_+)$, this would just say that $\Phi^\dagger$ transforms according to the representation of the gauge group that is contragredient to the representation furnished by Φ , and any function of Φ and $\Phi^\dagger$ that is invariant under global gauge transformations would also be invariant under local gauge transformations. Because x_+ and x_- are different, we must introduce a gauge connection matrix $\Gamma_{nm}(x, \theta)$, with the transformation property

$$\Gamma(x, \theta) \longrightarrow \exp \left(+i \sum_A t_A \Lambda^A(x_+)^* \right) \Gamma(x, \theta) \exp \left(-i \sum_A t_A \Lambda^A(x_+) \right). \quad (27.1.4)$$

Then by multiplying $\Phi^\dagger$ on the right with Γ we obtain a superfield that transforms as

$$\left[\Phi^\dagger(x, \theta) \Gamma(x, \theta) \right]_n \longrightarrow \sum_m \left[\Phi^\dagger(x, \theta) \Gamma(x, \theta) \right]_m \left[\exp \left(-i \sum_A t_A \Lambda^A(x_+) \right) \right]_{mn}, \quad (27.1.5)$$

so that any globally gauge-invariant function constructed from Φ and $\Phi^\dagger \Gamma$ (and not their derivatives or complex conjugates) will also be locally gauge-invariant. One obvious example is the gauge-invariant version $(\Phi^\dagger \Gamma \Phi)_D$ of the D -term in the Lagrangian constructed in Section 26.4.

Any $\Gamma(x, \theta)$ that transforms as in Eq. (27.1.4) will allow us to construct gauge-invariant Lagrangians of chiral superfields. The choice is not unique; if Γ transforms as in Eq. (27.1.4), and we multiply on the right with any left-chiral superfield Υ_L with the transformation rule

$$\Upsilon_L(x, \theta) \longrightarrow \exp \left(i \sum_A t_A \Lambda^A(x_+) \right) \Upsilon_L(x, \theta) \exp \left(-i \sum_A t_A \Lambda^A(x_+) \right),$$

then we obtain a new gauge connection that also satisfies Eq. (27.1.4). One simplification is to take $\Gamma(x, \theta)$ Hermitian:

$$\Gamma^\dagger(x, \theta) = \Gamma(x, \theta). \quad (27.1.6)$$

This is always possible if there is any $\Gamma(x, \theta)$ that satisfies Eq. (27.1.4), for then by taking the Hermitian adjoint of Eq. (27.1.4) we easily see that $\Gamma^\dagger(x, \theta)$ transforms in the same way as $\Gamma(x, \theta)$, so if $\Gamma(x, \theta)$ is not Hermitian then we can replace it with its Hermitian part $(\Gamma + \Gamma^\dagger)/2$ (or, if that vanishes, by its anti-Hermitian part $(\Gamma - \Gamma^\dagger)/2i$.) Another simplification of great physical importance is to express $\Gamma(x, \theta)$ in terms of fields whose gauge transformation properties do not depend on the specific representation t_A of the gauge algebra under which the chiral superfield $\Phi(x, \theta)$ transforms, so that these fields can be used to form a suitable matrix $\Gamma(x, \theta)$ for chiral superfields that transform according to any representation of the gauge group. For this purpose, it is useful to recall the Baker–Hausdorff formula, which states that, for arbitrary matrices a and b ,

$$e^a e^b = \exp \left(a + b + \frac{1}{2}[a, b] + \frac{1}{12}[a, [a, b]] + \frac{1}{12}[b, [b, a]] + \dots \right), \quad (27.1.7)$$

where ‘ $\dots$ ’ denotes higher-order terms that can be written as multiple commutators of a s and b s, like the second- and third-order terms that are shown explicitly. It follows from this

that for any representation of a Lie algebra, we have

$$\exp\left(\sum_A a^A t_A\right) \exp\left(\sum_A b^A t_A\right) = \exp\left(\sum_A f^A(a, b) t_A\right), \quad (27.1.8)$$

where

$$\begin{aligned} f^A(a, b) = & a^A + b^A + \frac{i}{2} \sum_{BC} C^A_{BC} a^B b^C - \frac{1}{12} \sum_{BCDE} C^A_{BC} C^C_{DE} a^B a^D b^E \\ & - \frac{1}{12} \sum_{BCDE} C^A_{BC} C^C_{DE} b^B b^D a^E + \dots, \end{aligned} \quad (27.1.9)$$

which depends on the Lie algebra through its structure constants C^A_{BC} , defined as usual by

$$[t_B, t_C] = i \sum_A C^A_{BC} t_A,$$

but does not depend on the particular representation furnished by the t_A . We will therefore take $\Gamma(x, \theta)$ in the form

$$\Gamma(x, \theta) = \exp\left(-2 \sum_A t_A V^A(x, \theta)\right), \quad (27.1.10)$$

where $V^A(x, \theta)$ are a set of real superfields (so that Γ is Hermitian), not depending on the representation of the gauge algebra furnished by the t_A .

We can achieve an important further simplification by noting an additional symmetry of supersymmetric gauge theories. If some function of Φ and $\Phi^\dagger \Gamma$ is invariant under global gauge transformations, then it will automatically be invariant not only under the local gauge transformations (27.1.2)–(27.1.4), but also under the larger group of *extended gauge transformations*

$$\Phi_{nL}(x, \theta) \longrightarrow \sum_m \left[\exp\left(i \sum_A t_A \Omega^A(x, \theta)\right) \right]_{nm} \Phi_{mL}(x, \theta) \quad (27.1.11)$$

and

$$\Gamma(x, \theta) \longrightarrow \exp\left(-i \sum_A t_A \Omega^A(x, \theta)\right) \Gamma(x, \theta) \exp\left(+i \sum_A t_A \Omega^A(x, \theta)^*\right), \quad (27.1.12)$$

where $\Omega^A(x, \theta)$ is an arbitrary left-chiral superfield—that is, an arbitrary function of θ_L as well as x_+ . Under this transformation,

$$V^A(x, \theta) \longrightarrow V^A(x, \theta) + \frac{i}{2} [\Omega^A(x, \theta) - \Omega^A(x, \theta)^*] + \dots, \quad (27.1.13)$$

where ‘ $\dots$ ’ denotes terms arising from the commutators in Eq. (27.1.7), which are of first or higher order in gauge coupling constants. As a general left-chiral superfield, Ω may be

written in the form (26.3.11)

$$\begin{aligned}\Omega^A(x, \theta) = & W^A(x) - \sqrt{2} \left(\bar{\theta} \frac{1 + \gamma_5}{2} w^A(x) \right) + \mathcal{W}^A(x) \left(\bar{\theta} \frac{1 + \gamma_5}{2} \theta \right) \\ & + \frac{1}{2} (\bar{\theta} \gamma_5 \gamma_\mu \theta) \partial^\mu W^A(x) - \frac{1}{\sqrt{2}} (\bar{\theta} \gamma_5 \theta) \left(\bar{\theta} \not{\partial} \frac{1 + \gamma_5}{2} w^A(x) \right) \\ & - \frac{1}{8} (\bar{\theta} \gamma_5 \theta)^2 \square W^A(x),\end{aligned}\tag{27.1.14}$$

in which $W^A(x)$ and $\mathcal{W}^A(x)$ are arbitrary complex functions of x^μ , and we introduced Majorana spinors $w^A(x)$ defined so that the left-handed spinor components of the superfields are $\frac{1}{2}(1 + \gamma_5)w^A(x)$. Using the complex conjugation properties (26.A.21) of Majorana bilinears, the complex conjugate of Eq. (27.1.14) gives

$$\begin{aligned}\Omega^A(x, \theta)^* = & W^{A*}(x) - \sqrt{2} \left(\bar{\theta} \frac{1 - \gamma_5}{2} w^A(x) \right) + \mathcal{W}^{A*}(x) \left(\bar{\theta} \frac{1 - \gamma_5}{2} \theta \right) \\ & - \frac{1}{2} (\bar{\theta} \gamma_5 \gamma_\mu \theta) \partial^\mu W^{A*}(x) - \frac{1}{\sqrt{2}} (\bar{\theta} \gamma_5 \theta) \left(\bar{\theta} \not{\partial} \frac{1 - \gamma_5}{2} w^A(x) \right) \\ & - \frac{1}{8} (\bar{\theta} \gamma_5 \theta)^2 \square W^{A*}(x).\end{aligned}\tag{27.1.15}$$

We write the real superfields $V^A(x, \theta)$ in terms of component fields as in Eq. (26.2.10):

$$\begin{aligned}V^A(x, \theta) = & C^A(x) - i(\bar{\theta} \gamma_5 \omega^A(x)) - \frac{i}{2} (\bar{\theta} \gamma_5 \theta) M^A(x) - \frac{1}{2} (\bar{\theta} \theta) N^A(x) \\ & + \frac{i}{2} (\bar{\theta} \gamma_5 \gamma^\mu \theta) V_\mu^A(x) - i(\bar{\theta} \gamma_5 \theta) \left(\bar{\theta} \left[\lambda^A(x) + \frac{1}{2} \not{\partial} \omega^A(x) \right] \right) \\ & - \frac{1}{4} (\bar{\theta} \gamma_5 \theta)^2 \left(D^A(x) + \frac{1}{2} \square C^A(x) \right),\end{aligned}\tag{27.1.16}$$

where $C^A(x)$, $M^A(x)$, $N^A(x)$, and $V_\mu^A(x)$ are all real, and $\omega^A(x)$ and $\lambda^A(x)$ are Majorana spinors. Using Eqs. (27.1.14)–(27.1.16) in Eq. (27.1.13), we find that the component fields of the gauge superfield undergo the extended gauge transformation

$$\begin{aligned}C^A(x) & \longrightarrow C^A(x) - \text{Im } W^A(x) + \dots, \\ \omega^A(x) & \longrightarrow \omega^A(x) + \frac{1}{\sqrt{2}} w^A(x) + \dots, \\ V_\mu^A(x) & \longrightarrow V_\mu^A(x) + \partial_\mu \text{Re } W^A(x) + \dots, \\ M^A(x) & \longrightarrow M^A(x) - \text{Re } \mathcal{W}^A(x) + \dots, \\ N^A(x) & \longrightarrow N^A(x) + \text{Im } \mathcal{W}^A(x) + \dots, \\ \lambda^A(x) & \longrightarrow \lambda^A(x) + \dots, \\ D^A(x) & \longrightarrow D^A(x) + \dots,\end{aligned}\tag{27.1.17}$$

where again ‘ $\dots$ ’ denotes terms that arise from the structure constants in Eq. (27.1.9) and that are therefore proportional to one or more factors of gauge coupling constants. We can

use such an extended gauge transformation to put the gauge superfields into a convenient form, known as *Wess–Zumino gauge* [1], in which

$$C^A(x) = \omega^A(x) = M^A(x) = N^A(x) = 0, \quad (27.1.18)$$

so that

$$V^A(x, \theta) = \frac{i}{2}(\bar{\theta}\gamma_5\gamma^\mu\theta)V_\mu^A(x) - i(\bar{\theta}\gamma_5\theta)(\bar{\theta}\lambda^A(x)) - \frac{1}{4}(\bar{\theta}\gamma_5\theta)^2 D^A(x). \quad (27.1.19)$$

To accomplish this to zeroth order in the coupling constants, it is only necessary to set $\text{Im } W^A(x) = C^A(x)$, $w^A(x) = -\sqrt{2}\omega^A(x)$, and $\mathcal{W}^A(x) = M^A(x) - iN^A(x)$. For Abelian gauge theories, in which the structure constants vanish, this ends our task. For non-Abelian gauge theories it is necessary to add terms to $\text{Im } W^A(x)$, $w^A(x)$, and $\mathcal{W}^A(x)$ of first order in gauge coupling constants to cancel the terms arising from commutators of zeroth-order terms, then add terms to $\text{Im } W^A(x)$, $w^A(x)$, and $\mathcal{W}^A(x)$ of second order in gauge coupling constants to cancel the terms arising from commutators of first-order terms with zeroth-order terms, and so on. It is not easy to calculate the series of terms in $\text{Im } W^A(x)$, $w^A(x)$, and $\mathcal{W}^A(x)$ needed to satisfy the gauge conditions (27.1.18) to all orders in gauge couplings, but there is no need—the important thing is that it is possible.

Inspection of the transformation rules (26.2.11)–(26.2.14) shows that the Wess–Zumino gauge condition (27.1.18) is not invariant under supersymmetry transformations unless $V_\mu^A = \lambda^A = 0$, and the condition $\lambda^A = 0$ is not supersymmetric unless also $D^A = 0$, in which case the whole superfield vanishes. Once we adopt Wess–Zumino gauge, the action is no longer invariant under either general extended gauge transformations or under supersymmetry, but it is invariant under supersymmetry transformations, which take us out of Wess–Zumino gauge, followed by suitable extended gauge transformations that take us back to Wess–Zumino gauge. (We will go into this explicitly in Section 27.8.) As we shall now see, it is also invariant under the ordinary gauge transformations (27.1.2)–(27.1.4), which preserve Wess–Zumino gauge.

With the gauge superfield satisfying the Wess–Zumino gauge condition (27.1.18), it becomes relatively easy to calculate its behavior under ordinary infinitesimal gauge transformations. In this case, $\Omega^A(x_+)$ are left-chiral superfields of the form (26.3.11), but with no ψ_L - or $\mathcal{F}$ -components, and with ϕ -components given by *real* infinitesimal functions $\Lambda^A(x)$:

$$\Omega^A(x_+) = \Lambda^A(x) + \frac{1}{2}(\bar{\theta}\gamma_5\gamma^\mu\theta)\partial_\mu\Lambda^A(x) - \frac{1}{8}(\bar{\theta}\gamma_5\theta)^2\Box\Lambda^A(x). \quad (27.1.20)$$

To calculate the product of exponentials in the transformation rule (27.1.4), we use a version of the Baker–Hausdorff formula:

$$\exp(a)\exp(X)\exp(b) = \exp[X + L_X \cdot (b - a) + (L_X \coth L_X) \cdot (b + a) + \cdots], \quad (27.1.21)$$

where a , b , and X are arbitrary matrices, L_X is the operator

$$L_X \cdot f = \frac{1}{2}[X, f], \quad (27.1.22)$$

and ‘...’ here denotes terms of second and higher order in a and/or b . In our case we have

$$\begin{aligned}
b + a &= 2 \sum_A t_A \operatorname{Im} \Lambda^A(x_+) = -i(\bar{\theta}\gamma_5\gamma_\mu\theta) \sum_A t_A \partial^\mu \Lambda^A(x), \\
b - a &= -2i \sum_A t_A \operatorname{Re} \Lambda^A(x_+) = -2i \sum_A t_A \left[\Lambda^A(x) - \frac{1}{8}(\bar{\theta}\gamma_5\theta)^2 \square \Lambda^A(x) \right], \\
X &= -2 \sum_A t_A V^A(x, \theta) \\
&= -2 \sum_A t_A \left[\frac{i}{2}(\bar{\theta}\gamma_5\gamma^\mu\theta) V_\mu^A(x) - i(\bar{\theta}\gamma_5\theta)(\bar{\theta}\lambda^A(x)) - \frac{1}{4}(\bar{\theta}\gamma_5\theta)^2 D^A(x) \right].
\end{aligned}$$

Now, every term in X involves at least one factor of θ_L and at least one factor of θ_R , while $a + b$ has just one factor of θ_L and one factor of θ_R , so we can drop any terms in L_X coth L_X of second or higher order in L_X . Since L_X coth L_X is an *even* function of L_X , this means that we can replace it with its term of zeroth order in L_X , which is just unity. Also, we may drop the term in $b - a$ proportional to $(\bar{\theta}\gamma_5\theta)^2$, since when acted on by L_X it would yield at least three factors of either θ_L or θ_R . Thus the argument of the exponential on the right-hand side of Eq. (27.1.21) may be replaced with

$$\begin{aligned}
X + \frac{1}{2}[X, b - a] + b + a &= -2 \sum_A t_A \left[V^A(x, \theta) \right. \\
&\quad + \sum_{BC} C^A_{BC} V^B(x, \theta) \Lambda^C(x) \\
&\quad \left. + \frac{i}{2}(\bar{\theta}\gamma_5\gamma_\mu\theta) \partial^\mu \Lambda^A(x) \right].
\end{aligned}$$

Thus for infinitesimal gauge transformations, the transformation rule (27.1.4) yields

$$V^A(x, \theta) \longrightarrow V^A(x, \theta) + \sum_{BC} C^A_{BC} V^B(x, \theta) \Lambda^C(x) + \frac{i}{2}(\bar{\theta}\gamma_5\gamma_\mu\theta) \partial^\mu \Lambda^A(x). \quad (27.1.23)$$

It is important to note that under ordinary gauge transformations, a gauge superfield in Wess–Zumino gauge remains in Wess–Zumino gauge. In terms of the component fields in Eq. (27.1.19), Eq. (27.1.23) reads

$$V_\mu^A(x) \longrightarrow \sum_{BC} C^A_{BC} V_\mu^B(x) \Lambda^C(x) + \partial_\mu \Lambda^A(x), \quad (27.1.24)$$

$$\lambda^A(x) \longrightarrow \sum_{BC} C^A_{BC} \lambda^B(x) \Lambda^C(x), \quad (27.1.25)$$

$$D^A(x) \longrightarrow \sum_{BC} C^A_{BC} D^B(x) \Lambda^C(x). \quad (27.1.26)$$

We recognize Eq. (27.1.24) as the usual Yang–Mills gauge transformation rule (15.1.9) of a gauge field, while Eqs. (27.1.25) and (27.1.26) tell us that the fields $\lambda^A(x)$ and $D^A(x)$ transform as ‘matter’ fields that belong to the adjoint representation of the gauge group.

The Majorana spinors λ^A are known as *gaugino* fields, while the real scalars D^A will turn out to be another set of auxiliary fields.

Next we must evaluate the matrix Γ that is needed in constructing gauge-invariant functions of chiral superfields. Because all terms with more than four factors of θ vanish, the expansion of the exponential is very simple in Wess–Zumino gauge:

$$\begin{aligned}\Gamma(x, \theta) &= \exp \left(-2 \sum_A t_A V^A(x, \theta) \right) \\ &= \mathbf{1} - i(\bar{\theta} \gamma_5 \gamma^\mu \theta) \sum_A t_A V_\mu^A(x) \\ &\quad - \frac{1}{2}(\bar{\theta} \gamma_5 \gamma^\mu \theta)(\bar{\theta} \gamma_5 \gamma^\nu \theta) \sum_{AB} t_A t_B V_\mu^A(x) V_\nu^B(x) \\ &\quad + 2i(\bar{\theta} \gamma_5 \theta) \sum_A t_A (\bar{\theta} \lambda^A(x)) + \frac{1}{2}(\bar{\theta} \gamma_5 \theta)^2 \sum_A t_A D^A(x).\end{aligned}$$

We can construct a gauge-invariant density by multiplying this on the right with a column vector of left-chiral superfields of the form (26.3.11):

$$\begin{aligned}\Phi_n(x, \theta) &= \phi_n(x) - \sqrt{2}(\bar{\theta} \psi_{nL}(x)) + \mathcal{F}_n(x) \left(\bar{\theta} \frac{1 + \gamma_5}{2} \theta \right) \\ &\quad + \frac{1}{2}(\bar{\theta} \gamma_5 \gamma_\mu \theta) \partial^\mu \phi_n(x) - \frac{1}{\sqrt{2}}(\bar{\theta} \gamma_5 \theta)(\bar{\theta} \not{\partial} \psi_{nL}(x)) - \frac{1}{8}(\bar{\theta} \gamma_5 \theta)^2 \square \phi_n(x),\end{aligned}$$

and multiplying on the left with the column

$$\begin{aligned}\Phi_n(x, \theta)^* &= \phi_n^*(x) - \sqrt{2}(\overline{\psi_{nL}(x)} \theta) + \mathcal{F}_n^*(x) \left(\bar{\theta} \frac{1 - \gamma_5}{2} \theta \right) \\ &\quad - \frac{1}{2}(\bar{\theta} \gamma_5 \gamma_\mu \theta) \partial^\mu \phi_n^*(x) - \frac{1}{\sqrt{2}}(\bar{\theta} \gamma_5 \theta) \partial_\mu (\overline{\psi_{nL}(x)} \gamma^\mu \theta) \\ &\quad - \frac{1}{8}(\bar{\theta} \gamma_5 \theta)^2 \square \phi_n^*(x).\end{aligned}$$

The term in this product of fourth order in θ is

$$\begin{aligned}
\left[\Phi^\dagger \Gamma \Phi\right]_{\theta^4} = & -\frac{1}{8}(\bar{\theta}\gamma_5\theta)^2 \left\{ [\phi^\dagger \square \phi] + [(\square \phi^\dagger)\phi] \right\} \\
& + (\bar{\theta}\gamma_5\theta) \left\{ [(\bar{\psi}_L\theta)(\bar{\theta}\gamma^\mu\partial_\mu\psi_L)] + [((\partial_\mu\bar{\psi}_L)\gamma^\mu\theta)(\bar{\theta}\psi_L)] \right\} \\
& + \frac{1}{4}(\bar{\theta}(1-\gamma_5)\theta)(\bar{\theta}(1+\gamma_5)\theta)[\mathcal{F}^\dagger\mathcal{F}] \\
& - \frac{1}{4}(\bar{\theta}\gamma_5\gamma^\mu\theta)(\bar{\theta}\gamma_5\gamma^\nu\theta)[\partial_\mu\phi^\dagger\partial_\nu\phi] \\
& - \frac{i}{2}(\bar{\theta}\gamma_5\gamma^\mu\theta)(\bar{\theta}\gamma_5\gamma^\nu\theta)\sum_A V_\mu^A \left\{ [\phi^\dagger t_A\partial_\nu\phi] - [(\partial_\nu\phi^\dagger)t_A\phi] \right\} \\
& - \frac{1}{2}(\bar{\theta}\gamma_5\gamma^\mu\theta)(\bar{\theta}\gamma_5\gamma^\nu\theta)\sum_{AB} V_\mu^A V_\nu^B [\phi^\dagger t_A t_B \phi] \\
& - 2i(\bar{\theta}\gamma_5\gamma_\mu\theta)\sum_A V^{A\mu}[(\bar{\psi}_L\theta)t_A(\bar{\theta}\psi_L)] \\
& - 2i\sqrt{2}(\bar{\theta}\gamma_5\theta)\sum_A [(\bar{\psi}_L\theta)t_A(\bar{\theta}\lambda^A)\phi] \\
& - 2i\sqrt{2}(\bar{\theta}\gamma_5\theta)\sum_A [\phi^\dagger(\bar{\lambda}^A\theta)t_A(\bar{\theta}\psi_L)] \\
& + \frac{1}{2}(\bar{\theta}\gamma_5\theta)^2\sum_A D_A[\phi^\dagger t_A\phi],
\end{aligned}$$

in which we use square brackets to indicate scalar products in the flavor indices n, m , and continue to use round brackets to indicate scalar products in Dirac indices. Just as in Section 26.4, we may use the identities (26.A.17)–(26.A.19) to put all θ dependence in the form of an over-all factor $(\bar{\theta}\gamma_5\theta)^2$:

$$\begin{aligned}
\left[\Phi^\dagger \Gamma \Phi\right]_{\theta^4} = & (\bar{\theta}\gamma_5\theta)^2 \left\{ -\frac{1}{8}[\phi^\dagger \square \phi] - \frac{1}{8}[(\square \phi^\dagger)\phi] \right. \\
& + \frac{1}{4}[(\bar{\psi}_L\gamma^\mu\partial_\mu\psi_L)] - \frac{1}{4}[(\partial_\mu\bar{\psi}_L)\gamma^\mu\psi_L] \\
& - \frac{1}{2}[\mathcal{F}^\dagger\mathcal{F}] + \frac{1}{4}[\partial_\mu\phi^\dagger\partial^\mu\phi] \\
& + \frac{i}{2}\sum_A V_\mu^A[\phi^\dagger t_A\partial^\mu\phi] - \frac{i}{2}\sum_A V_\mu^A[(\partial^\mu\phi^\dagger)t_A\phi] \\
& + \frac{1}{2}\sum_{AB} V_\mu^A V^{B\mu}[\phi^\dagger t_A t_B \phi] - \frac{i}{2}\sum_A V_\mu^A[(\bar{\psi}_L\gamma^\mu t_A\psi_L)] \\
& - \frac{i}{\sqrt{2}}\sum_A [(\bar{\psi}_L t_A \lambda^A)\phi] + \frac{i}{\sqrt{2}}\sum_A [\phi^\dagger(\bar{\lambda}^A t_A \psi_L)] \\
& \left. + \frac{1}{2}\sum_A D_A[\phi^\dagger t_A\phi] \right\}.
\end{aligned}$$

The D -term is the coefficient of $-\frac{1}{4}(\bar{\theta}\gamma_5\theta)^2$ minus $\frac{1}{2}\square$ acting on the θ -independent term,

which for $[\Phi^\dagger \Gamma \Phi]$ is $[\phi^\dagger \phi]$, so

$$\begin{aligned}
[\Phi^\dagger \Gamma \Phi]_D = & -2[\partial_\mu \phi^\dagger \partial^\mu \phi] - [(\overline{\psi_L} \gamma^\mu \partial_\mu \psi_L)] + [(\partial_\mu \overline{\psi_L}) \gamma^\mu \psi_L] + 2[\mathcal{F}^\dagger \mathcal{F}] \\
& - 2i \sum_A V_\mu^A [\phi^\dagger t_A \partial^\mu \phi] + 2i \sum_A V_\mu^A [(\partial^\mu \phi^\dagger) t_A \phi] \\
& - 2 \sum_{AB} V_\mu^A V^{B\mu} [\phi^\dagger t_A t_B \phi] + 2i \sum_A V_\mu^A [(\overline{\psi_L} \gamma^\mu t_A \psi_L)] \\
& + 2i\sqrt{2} \sum_A [(\overline{\psi_L} t_A \lambda^A) \phi] - 2i\sqrt{2} \sum_A [\phi^\dagger (\overline{\lambda^A} t_A \psi_L)] \\
& - 2 \sum_A D_A [\phi^\dagger t_A \phi].
\end{aligned}$$

To see that this is gauge-invariant, we note that it may be written as

$$\begin{aligned}
\frac{1}{2}[\Phi^\dagger \Gamma \Phi]_D = & -[(D_\mu \phi)^\dagger D^\mu \phi] - \frac{1}{2}[(\overline{\psi_L} \gamma^\mu D_\mu \psi_L)] + \frac{1}{2}[(\overline{D_\mu \psi_L} \gamma^\mu \psi_L)] + [\mathcal{F}^\dagger \mathcal{F}] \\
& + i\sqrt{2} \sum_A [(\overline{\psi_L} t_A \lambda^A) \phi] - i\sqrt{2} \sum_A [\phi^\dagger (\overline{\lambda^A} t_A \psi_L)] - \sum_A D_A [\phi^\dagger t_A \phi],
\end{aligned} \tag{27.1.27}$$

where D_μ is the gauge-invariant derivative (15.1.10):

$$D_\mu \psi_L \equiv \partial_\mu \psi_L - i \sum_A t_A V_\mu^A \psi_L, \quad D_\mu \phi \equiv \partial_\mu \phi - i \sum_A t_A V_\mu^A \phi. \tag{27.1.28}$$

Eq. (27.1.27) is thus a suitable gauge-invariant kinematic Lagrangian for the scalar and spinor components of a left-chiral superfield, now supplemented with Yukawa couplings of gaugino fields to the scalar and spinor components of chiral superfields as well as terms involving auxiliary fields $\mathcal{F}_n$ and D_A .